Lab 6 - Move, Copy, Rotate and Offset

Course: WSQ Technical Drawing with AutoCAD (TGS-2021005538) | Topic 03: Modifying Objects

Objective: LO2/LO3 - modify 2D drawings efficiently to build up a design (A3, K3)

Goal

Build practice drawing EX-4 by drawing one feature and then positioning the rest with MOVE, COPY (multiple), ROTATE with reference angle and OFFSET for the parallel profiles.

You'll produce

The completed EX-4 drawing produced with modify commands rather than re-drawing each feature.

Commands / tools: MOVE, COPY (Multiple/Array), ROTATE (Reference), OFFSET

Files in this lab

- ex4.dwg - AutoCAD drawing file
- ex4-drawing-sheet.pdf - drawing sheet / reference (PDF)
- ex4.png - reference image

Step-by-step

1. Open the EX-4 sheet (ex4-drawing-sheet.pdf) and spot the repeated features
2. Draw the base profile, then offset it to create the parallel inner/outer edges

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O → distance → pick side
```

3. Draw one hole feature and copy it to every dimensioned centre

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CO → base point → second points
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4. Move a mis-placed feature precisely from its centre to the correct intersection

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M → base → destination
```

5. Rotate the angled feature about its base point, using Reference to match the sheet angle

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RO → base → R → ref angle → new angle
```

6. Check against ex4.dwg and save your version

```
SAVEAS ex4-mine.dwg
```

Check your work

Feature spacings measure correct with DIST, offsets are at the specified distances, and the rotated feature sits at exactly the sheet's angle.